

CHAPTER 18

Higher-Order Difference Equations

$$\Delta y_t = y_{t+1} - y_t$$

Strictly speaking, a second-order Difference Equation is one that involves an expression $\Delta^2 y_t$, called the second difference of y_t , but contains no difference of order higher than 2.

The symbol Δ^2 , the discrete-time counterpart of the symbol d^2/dt^2 , is an instruction to "take the second difference" as follows:

$$\Delta y_t = ?$$

$$\Delta^2 y_t = ?$$

$$\Delta^2 y_t = \Delta(\Delta y_t) = \Delta(y_{t+1} - y_t) \\ = \Delta y_{t+1} - \Delta y_t$$

$$y_{t+2} - y_{t+1} - (y_{t+1} - y_t) \\ y_{t+2} - y_{t+1} - y_{t+1} + y_t \\ y_{t+2} - 2y_{t+1} + y_t$$

$$\Delta^2 y_t = y_{t+2} - 2y_{t+1} + y_t$$

Second-Order Linear Difference Equations with Constant Coefficients and Constant Term:

A simply variety of second-order difference equations takes the form

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = c$$

Constant Co-efficient: $(a, a_1), (y, y_1)$

Constant Term: c

y_p = Non-homogeneous equation

y_c = Homogeneous equation

Particular Solution:

$$y_p = \frac{c}{1+a_1+a_2} \quad (\text{case of } a_1+a_2 \neq -1)$$

Example,

Find the particular integral

of $y_{t+2} - 3y_{t+1} + 4y_t = 6$

$$a_1 = -3, a_2 = 4, c = 6$$

$$y_p = \frac{c}{1+a_1+a_2}$$

$$= \frac{6}{1+(-3)+4}$$

$$= \frac{6}{1-3+4}$$

$$= \frac{6^3}{2_1}$$

$$\boxed{y_p = 3}$$

Thus we can rewrite the particular solution as

$$y_p = \frac{c}{a_1 + 2} t \quad (\text{case of } a_1 + a_2 = -1; a_1 \neq -2)$$

Example:

Find the particular solution

$$y_{t+2} + y_{t+1} - 2y_t = 12$$

Here,

$$a_1 = 1, a_2 = -2, c = 12$$

$$y_p = \frac{c}{a_1 + 2} t$$

$$y_p = \frac{12}{1+2} t$$

$$y_p = \frac{12}{3} t$$

$$\boxed{y_p = 4t}$$

This particular solution represents a moving equilibrium.

As you may verify, in this case the particular solution turns out to be

$$y_p = kt^2 = \frac{c}{2} t^2$$

(case of $a_1 = -2$; $a_2 = 1$)

Complementing Function

$$y_{t+2} + a_1 y_{t+1} + a_2 y_t = 0 \quad \text{--- eq. (i)}$$

Upon substitution of the trial solution into eq. (i), the equation becomes,

$$Ab^{t+2} + a_1 Ab^{t+1} + a_2 Ab^t = 0$$

after cancelling the (nonzero) common factors Ab^t

$$b^2 + a_1 b + a_2 = 0$$

Quadratic equation:

Two characteristics roots:

$$b_1, b_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}$$

Case 1:

(Distinct real roots)

$$y_c = A_1 b_1^t + A_2 b_2^t$$

Case 2:

(repeated ^{real} root)

$$b (= b_1 = b_2) = -\frac{a_1}{2}$$

The complementary function for the repeated real roots case is therefore,

$$y_c = A_3 b^t + A_4 t b^t$$

Example:

$$y_{t+2} + y_{t+1} - 2y_t = 12$$

$$a_1 = 1, a_2 = -2, c = 12$$

$$b_1 = 1, b_2 = -2 \rightarrow \text{find value}$$

Quadratic equation

OR

Complementary Function: Synthetic Division

$$y_c = A_1 b_1^t + A_2 b_2^t$$

$$y_c = A_1 (1)^t + A_2 (-2)^t$$

$$\boxed{y_c = A_1 + A_2 (-2)^t}$$

$$y_p = \frac{c}{a_1 + 2} t = \frac{12}{1 + 2} t$$

$$y_p = \frac{12}{1 + 2} t = \frac{12}{3} t$$

$$\boxed{y_p = 4t}$$

General solution:

$$y_t = y_c + y_p$$

$$y_t = A_1 + A_2 (-2)^t + 4t$$

Example:

$$y_{t+2} + 6y_{t+1} + 9y_t = 4$$

(Repeated Real Roots)

Method

$$a_1^2 = 4a_2 \Rightarrow 6^2 = 4 \times 9 \Rightarrow 36 = 36 \checkmark$$

$$a_1 = 6, a_2 = 9, c = 4$$

$$b_1 = b_2 = -3 \rightarrow \text{find: } b_1, b_2 = -\frac{a_1}{2}$$

$$\Rightarrow y_c = ?$$

$$= -\frac{6^2}{2}$$

$$= -3$$

$$y_c = A_3 b^t + A_4 t b^t$$

$$y_c = A_3 (-3)^t + A_4 t (-3)^t$$

$$\Rightarrow y_p = ?$$

$$y_p = \frac{c}{1+a_1+a_2} = \frac{4}{1+6+9}$$

$$y_p = \frac{4}{16} = \boxed{\frac{1}{4}}$$

General solution:

$$y_t = y_c + y_p$$

$$y_t = A_3(-3)^t + A_4(-3)^t + 1/4$$

Case 3

(complex roots)

$$b_1, b_2 = h \pm vi$$

$$h = \frac{-a_1}{2}$$

$$v = \frac{\sqrt{4a_2 - a_1^2}}{2}$$

Complementary Function

$$y_c = A_1(h+vi)^t + A_2(h-vi)^t$$

Theorem,

$$(h \pm vi)^t = R^t (\cos \theta t \pm i \sin \theta t)$$

$$R = \sqrt{h^2 + v^2}$$

and θ is the radian measure

of the angle in the interval $(0, 2\pi)$, which satisfies the condition

$$\cos \theta = \frac{h}{R} = \frac{-a_1}{2\sqrt{a_2}}$$

$$\sin \theta = \frac{v}{R} = \sqrt{1 - \frac{a_1^2}{4a_2}}$$

Example,

$$y^2 + 2 + \frac{1}{4}y^2 = 5$$

$$a_1 = 0, a_2 = \frac{1}{4}, c = 5$$

$$h = \frac{-a_1}{2}, \quad v = \frac{\sqrt{4a_2 - a_1^2}}{2}$$

$$h = \frac{-0}{2}, \quad v = \frac{\sqrt{4(1/4) - (0)^2}}{2}$$

$$\boxed{h = 0}, \quad \boxed{v = \frac{1}{2}}$$

$$b_1, b_2 = h + vi$$

$$\boxed{b_1, b_2 = 0 + \frac{1}{2}i}$$

$$R = \sqrt{h^2 + V^2}$$

$$R = \sqrt{(0)^2 + (1/2)^2}$$

$$\boxed{R = 1/2}$$

$$\cos \theta = \frac{h}{R} = \frac{0}{1/2} = \underline{0}$$

$$\sin \theta = \frac{V}{R} = \frac{1/2}{1/2} = \frac{1}{1} = \underline{1}$$

$$y_c = ?$$

$$y_c = R^t (A_5 \cos \theta t + A_6 \sin \theta t)$$

$$\boxed{y_c = (1/2)^t (A_5 \cos \pi/2 t + A_6 \sin \pi/2 t)}$$

$$y_p = ?$$

$$y_p = \frac{C}{1 + G_1 + G_2} = \frac{5}{1 + 0 + 1/4}$$

$$y_p = \frac{5}{1 + 1/4}$$

$$y_p = \frac{5}{\frac{4+1}{4}}$$

$$y_p = \frac{5}{5/4}$$

$$\boxed{y_p = 4}$$

General Solution:

$$y_b = y_c + y_p$$

$$y_b = \left(\frac{1}{2}\right)^t \left(A_5 \cos \frac{\pi t}{2} + A_6 \sin \frac{\pi t}{6} \right) + 4 \quad \underline{\underline{\text{Ans}}}$$

The Convergence of the
Time path

A time path will be convergent—
whatever the initial conditions
may be — if and only if the
dominant root is less than 1

in absolute value.

Exercise 18.1

Q: 1

Write out the characteristics
for each of the following
and find the characteristic
roots:

(a) $y_{t+2} - y_{t+1} + \frac{1}{2} y_t = 2$